$$\mathrm{f}(x) = \frac{8x - 5}{(2x - 1)(4x - 3)} \qquad x > 1$$

(a) Express $\mathrm{f}(x)$ in partial fractions.

(3)

(b) Hence find $\int \mathrm{f}(x)\,\mathrm{d}x$

(3)

(c) Use the answer to part (b) to find the value of k for which

$$\int_{k}^{3k} \mathrm{f}(x)\,\mathrm{d}x = \frac{1}{2}\ln 20$$

(5)